

Photosynthesis and Cellular Respiration - Study Notes

Biology 101, Week 4. Key definitions, formulas and processes.

1. Core Definitions

Photosynthesis: the process by which green plants, algae and some bacteria convert light energy into chemical energy stored in glucose.

Chlorophyll: the green pigment in chloroplasts that absorbs light, mainly in the blue and red wavelengths; reflects green.

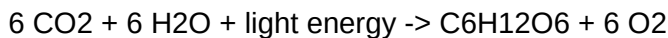
Stomata: pores on the leaf underside that regulate gas exchange.

ATP (adenosine triphosphate): the cell's main energy currency.

NADPH: electron carrier produced in the light reactions.

Calvin cycle: the light-independent reactions that fix CO₂ into sugar.

2. The Master Equation



Carbon dioxide + water, powered by light, yield glucose and oxygen.

3. Two Stages of Photosynthesis

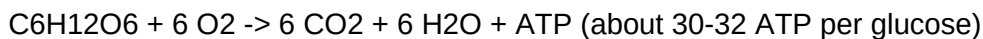
Stage A - Light reactions (thylakoid membranes):

- Light splits water (photolysis), releasing O₂.
- Electron transport chain builds ATP and NADPH.

Stage B - Calvin cycle (stroma):

- Carbon fixation: RuBisCO attaches CO₂ to RuBP.
- Reduction: ATP and NADPH convert 3-PGA into G3P.
- Regeneration: RuBP is rebuilt so the cycle can continue.

4. Cellular Respiration - the mirror process



Glycolysis (cytoplasm) → Krebs cycle (mitochondrial matrix) → oxidative phosphorylation (inner mitochondrial membrane).

5. Frequently Confused Pairs

Chlorophyll vs chloroplast: pigment vs organelle.

Light reactions vs Calvin cycle: energy capture vs carbon fixation.

Aerobic vs anaerobic respiration: with vs without oxygen.

6. Practice Questions

Q1. Where exactly does the Calvin cycle take place?

Q2. Which molecule is both a reactant of respiration and a product of photosynthesis? (Hint: there are two correct answers.)

Q3. Explain why the light reactions stop within seconds in the dark while the Calvin cycle can continue briefly.